LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to surface finish, meaning of surface texture and their definition.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to surface finish, meaning of surface texture and their definition.
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Nil
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is Engineering Metrology?
(b) Why is surface finish important in engineering?
(c) Can you define the term 'surface texture'?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Engineering Metrology • Definition of Surface Finish • Importance of Surface Finish • Factors affecting Surface Finish • Applications in Aeronautical Engineering	L & Ds	PPT, IB	15 min
(b)	• Meaning of Surface Texture • Components of Surface Texture • Difference between Roughness and Waviness • Role of Lay in Surface Texture • Standardization of Surface Texture	L & Ds	PPT, PDF	20 min
(c)	• Definition of Surface Roughness • Definition of Surface Waviness • Definition of Surface Lay • Definition of Surface Flaws • Summary of Surface Texture Parameters	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between surface roughness and surface waviness.
(b) How does surface finish affect the performance of aircraft components?
(c) List the main components that make up surface texture.

12. Home Assignment and Follow Up	Prepare a brief note on how surface finish requirements differ between engine parts and structural airframe components, referencing the next lesson plan on surface irregularities.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Surface roughness, meaning of real surface, geometrical surface, effective surface, and various irregularities.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min